

Prarthana Krishnamurthy

krishnamurthy.prarth@gmail.com | [github/pkrishna1801](https://github.com/pkrishna1801) | [linkedin/prarthana181k](https://www.linkedin.com/in/prarthana181k) | pkrishna1801.github.io | 857-396-4456

Summary

Machine learning engineer with hands-on experience building ML and computer vision models for scientific R&D, spanning molecular property prediction, drug discovery (QSAR, SHAP interpretability, conformal prediction), and medical imaging (segmentation, synthesis). Skilled in Python, PyTorch, and scikit-learn, with a track record of collaborating with scientists to turn research data into decision-ready models.

Experience

Machine Learning Intern – Molecular Modeling

May 2026 – Jul 2026

Alkermes Inc.

Waltham, MA

- Developed molecular property models (Chemprop, PyTorch, scikit-learn) to support lead optimization, achieving an average R^2 of 0.7 and Spearman correlation of 0.73 for compound prioritization in DMTA cycles
- Boosted model performance 28.5% by applying Bayesian hyperparameter optimization (Optuna)
- Applied SHAP to MACCS-fingerprint models across multiple endpoints (potency, permeability, hERG, solubility), identifying the substructures driving each prediction and delivering interpretable structure-activity signals
- Integrated conformal prediction to property models, generating calibrated uncertainty estimates to flag high-confidence compounds for testing
- Presented new ML products and services to the VP of Research and stakeholders, driving adoption across 3 teams and 20+ users

Machine Learning Research Assistant

Jan 2025 – Apr 2026

Northeastern University (Advisor: Prof. Dino Konstantopoulos)

Boston, MA

- Architected transformer and diffusion-based generative models, leveraging flow matching to address complex non-autoregressive text generation constraints
- Scaled training across 8x NVIDIA H100s via PyTorch FSDP and NCCL optimization, sustaining 95%+ GPU utilization
- Reduced diffusion validation loss by 92.3%, achieving a 0.93 BERT semantic similarity score
- Resolved sampling error accumulation by implementing DPM-Solver++, reducing discretization error to 0.65
- Delivered SOTA performance metrics including BLEU-4: 0.267 and ROUGE-L: 0.742, demonstrating end-to-end ownership

Software Engineer, Data

Sep 2021 – Dec 2023

Societe Generale Global Solution Centre

Bangalore, KA

- Optimized SQL-based ETL pipelines to automate raw data ingestion from Azure Data Lake, delivering high-quality, structured datasets
- Partnered with data scientists to design Parquet schemas, unifying multi-format data from multiple sources within a central data lake
- Engineered PySpark transformation pipelines for raw-to-silver layer processing, performing large-scale data wrangling, deduplication, schema enforcement, and standardization
- Refactored 50+ legacy scripts into Airflow DAGs, enabling scalable workflow orchestration and centralized monitoring
- Productionized 3+ Dockerized enterprise applications on Azure Kubernetes Service, processing 1M+ daily transactions

Software Engineering Intern

Mar 2021 – May 2021

Capgemini

Bangalore, KA

- Designed and built 15+ RESTful microservices using Java 17 and Spring Boot, reducing API latency by 35% through optimized Hibernate JPA caching
- Implemented scalable multi-tenant microservices, ensuring high availability and data consistency for enterprise clients
- Developed RESTful APIs, optimizing response times through efficient data handling and database query optimization
- Engineered unit and integration test suites (JUnit, Mockito) to ensure API reliability before deployment
- Leveraged clean code principles (SOLID, DRY) and refactoring practices to improve codebase maintainability and readability

Publications

- [Non-Autoregressive Text Generation Using Diffusion Models](#), Master's Thesis, Northeastern University

Work Sample

Property-Conditioned Small Molecule Design | *RDKit, GPT-2, PyTorch*

[\[Code\]](#)

- Developed a conditional GPT-2 model in PyTorch to generate novel drug-like molecules from a 1.9M+ compound dataset
- Used RDKit to compute molecular descriptors and condition generation on target properties (LogP, QED)

Object Detection MLOps Pipeline | *AWS ECS, Docker, CodeBuild, Terraform*

[\[Code\]](#)

- Deployed scalable MLOps microservices (React, YOLO, DepthAnything) on AWS ECS with ALB and autoscaling

- Built a CI/CD pipeline (CodeBuild, ECR, GitHub webhooks) on Terraform-provisioned infrastructure for automated deployments

Bidirectional MRI Synthesis | *DDPM, U-Net*

[\[Code\]](#)

- Engineered a DDPM-based U-Net architecture for bidirectional T1-to-T2 MRI image translation
- Achieved an average PSNR of 23.8 and SSIM of 0.81 on synthesized MRI outputs

Distributed training for Liver Tumor Segmentation | *HPC, Dask, DDP, U-Net*

[\[Code\]](#)

- Implemented data parallelism for UNet-based liver tumor segmentation across 4 GPUs, achieving a 1.83x speedup
- Implemented model parallelism across 3 GPUs using Dask and DDP, achieving a 1.41x speedup

Adaptive Image Processing | *GoogleNet, CLAHE, Image Filtering*

[\[Code\]](#)

- Trained a GoogleNet-based classifier for high-accuracy image noise-type detection
- Applied adaptive preprocessing and filtering (CLAHE, OpenCV) to improve image quality and automated feature extraction workflows

Technical Skills

Programming Languages: Python, SQL, Bash

ML / Computer Vision: PyTorch, TensorFlow, scikit-learn, OpenCV, Diffusion Models, Distributed Training, Optuna (Bayesian Optimization), SHAP (Explainable AI)

Cheminformatics: RDKit, Chemprop, OpenEye, DeepChem

Cloud & Tools: AWS, Azure, MLflow, Docker, Git

Education

Northeastern University, MS in Computer Software Engineering (GPA: 3.84/4.0)

Jan 2024 – Dec 2025

Visvesvaraya Technological University, BE in Electronics and Communication

Aug 2017 – Aug 2021